

ASR + TTS

The interface war is shifting from keyboards to microphones. Bhashini handles 50 crore speech requests a month. Build voice or lose the next 500 million users.

When did you last speak to your phone in Hindi? What worked? What didn't?

L2

WHAT'S INSIDE

- 01** ASR pipeline — five stages
- 02** Production ASR — 4 options
- 03** TTS — three generations
- 04** 20-line bilingual ASR
- 05** Production considerations

WHY THIS LESSON MATTERS

Half of India prefers voice. GPay processed ₹4.2 lakh crore via voice in FY25.

NUMBER 1

50 crore

From real-world deployments.

NUMBER 2

500 million

From real-world deployments.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

ASR pipeline stages.

2

Use Whisper / Deepgram / Bhashini.

3

Compare TTS families.

4

Hindi-English ASR demo in 20 lines.

5

Handle diarisation + streaming.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

ASR pipeline — five stages

Audio to text, step by step

02

Production ASR — 4 options

English-only → Whisper. Indic → Bhashini. Real-time → Deepgram.

03

TTS — three generations

Voice cloning brings new fraud risks
— ₹280 crore in FY25 voice-fraud attempts at Indian banks.

04

20-line bilingual ASR

Whisper large-v3 auto-detect

01

PART 1 OF 4

ASR pipeline — five stages

Audio to text, step by step

CONCEPT 1 OF 4

ASR pipeline — five stages

Audio input (16 kHz mono).

Feature extraction (MFCC / mel-spectrogram).

Acoustic model (Transformer / Conformer).

Language model (re-score outputs).

Decoding (beam search; streaming for real-time).

PIPELINE

5 stages

All hidden behind one API call today.

DEFINITION

ASR pipeline — five stages

Audio to text, step by step

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Production ASR — 4 options

English-only → Whisper. Indic → Bhashini. Real-time → Deepgram.

KEY POINTS

- English-only → Whisper. Indic → Bhashini. Real-time → Deepgram
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

CONCEPT 3 OF 4

TTS — three generations

Voice cloning brings new fraud risks — ₹280 crore in FY25 voice-fraud attempts at Indian banks.

DEFINITION

TTS — three generations

Voice cloning brings new fraud risks — ₹280 crore in FY25 voice-fraud attempts at Indian banks.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

20-line bilingual ASR

```
model = whisper.load_model("large-v3")  
result = model.transcribe("call.mp3", language=None)  
Auto-detects Hindi, English, or code-switched Hinglish.  
Returns segments with timestamps — great for diarisation.
```

SHIP

20 LOC

Production-grade Hinglish in one evening.

KEY POINTS

- `model = whisper.load_model("large-v3")`
- `result = model.transcribe("call.mp3", language=None)`
- Auto-detects Hindi, English, or code-switched Hinglish
- Returns segments with timestamps — great for diarisation

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Transcribe your own voice in Hinglish

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

What's one voice-first product your grandparent would use?

Q2

Where does your current product fall back to text?

Q3

Which Indic language matters most for your target users?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Standard ASR input rate:

A 8 kHz

B 16 kHz

C 44.1 kHz

D 48 kHz

Standard ASR input rate:



CORRECT ANSWER

16 kHz

WHY

16 kHz mono PCM is production standard

Language model in ASR:

A

Replaces acoustic

B

Re-scores outputs

C

Does nothing

D

Creates audio

Language model in ASR:

CORRECT ANSWER



Re-scores outputs

WHY

LM fixes homophones; picks grammatically likely phrases

Bhashini sponsor:

A

Reliance

B

MeitY

C

Google

D

IIT

Bhashini sponsor:



CORRECT ANSWER

MeitY

WHY

Ministry of Electronics and IT — Indian government

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Audio input (16 kHz mono).
Feature extraction (MFCC / mel-spectrogram).
Acoustic model (Transformer / Conformer).
Language model (re-score outputs).
Decoding (beam search; streaming for real-time).

PIPELINE

5 stages

All hidden behind one API call today.

THE TAKEAWAY

"Build accordingly."

WHAT TO NOTICE

1

ASR pipeline — five stages

2

Production ASR — 4 options

3

TTS — three generations

4

20-line bilingual ASR

Hinglish transcription compare

5-minute mixed call; Whisper vs Bhashini.

DELIVERABLES

- Record.
- Transcribe both.
- Score word-error-rate.
- Post comparison.

WHAT 'GOOD' LOOKS LIKE

Deliverable: WER table on forum.

ONE THING TO REMEMBER

“

"Build accordingly."

— AI Learning Hub

END OF LESSON 7

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 3 — Reinforcement Learning.